

ANAKTUVUK PASS, AK, USA

WMO: 701625

Lat: 68.135N

Lon: 151.742W

Elev: 2100

StdP: 13.61

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26542

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.9	(c) -33.1	(d) -30.1	(e) 1.1	(f) -19.0	(g) -29.1	(h) 1.2	(i) -19.0	(j) 26.8	(k) -1.7	(l) 24.9	(m) -5.5	(n) 7.5	(o) 30	(p) 0.912

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	70.9	55.1	66.7	53.6	63.8	52.1	56.7	67.5	55.0	64.8	53.3	61.9	9.2	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.2	62.6	59.8	49.9	57.4	58.3	47.9	53.3	57.5	25.0	67.6	23.9	64.3	22.9	62.6	63.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Max
(a) 22.1	(b) 19.7	(c) 18.1	(d) DB	(e) -40.6	(f) 75.0	(g) 4.3	(h) 2.2	(i) -43.7	(j) 76.6	(k) -46.3	(l) 77.9	(m) -48.7	(n) 79.1	(o) -51.8	(p) 80.7	(q) N/A
(a) 22.1	(b) 19.7	(c) 18.1	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	-8.3	-4.7	-2.1	11.2	32.9	49.6	53.3	45.3	34.1	13.7	0.0	-6.3
	DBStd	25.45	15.53	15.53	13.84	13.43	11.95	7.45	5.93	7.01	9.29	11.85	12.35	14.02
	HDD50	11804	1806	1531	1616	1164	532	93	39	176	477	1125	1501	1744
	HDD65	17024	2271	1951	2081	1614	995	462	364	610	926	1590	1951	2209
	CDD50	255	0	0	0	0	2	81	140	31	1	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	20	0	0	0	0	0	7	12	1	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	8.7	9.7	9.5	9.4	8.5	8.5	8.5	8.0	8.3	7.7	8.1	9.0	9.4
Precipitation	PrecAvg	11.9	0.7	0.7	0.5	0.4	0.6	1.2	1.8	2.0	1.4	1.1	0.8	0.8
	PrecMax	16.7	2.0	1.9	1.2	1.8	1.5	2.9	3.5	5.0	4.1	2.3	2.2	3.0
	PrecMin	7.4	0.1	0.0	0.1	0.0	0.2	0.4	0.8	0.7	0.3	0.2	0.0	0.1
	PrecStd	2.2	0.5	0.5	0.3	0.4	0.3	0.5	0.6	0.9	0.9	0.4	0.6	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.1	30.3	32.5	43.1	64.1	74.6	75.6	71.4	58.1	45.2	30.1	27.8
		MCWB	25.1	27.4	29.3	36.4	49.0	55.0	57.9	56.4	46.4	38.1	27.8	25.9
	2%	DB	24.3	27.0	27.3	37.2	59.0	70.4	71.8	64.0	54.4	36.8	27.0	23.0
		MCWB	22.3	24.4	24.8	32.2	46.5	53.6	56.5	52.5	45.0	33.6	24.6	21.1
	5%	DB	19.0	21.0	21.8	34.5	54.5	66.3	68.3	60.9	51.6	33.9	22.8	18.5
		MCWB	17.5	18.9	19.6	30.6	44.1	51.9	55.0	50.7	44.0	31.2	20.8	17.0
	10%	DB	15.7	17.3	17.5	30.5	50.3	63.5	64.5	56.8	47.5	30.0	18.2	14.0
		MCWB	14.3	15.7	15.8	27.4	42.0	50.7	53.6	48.7	42.2	28.1	16.6	12.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.7	27.9	29.7	36.7	50.4	57.4	59.8	57.1	48.4	39.1	28.2	26.3
		MCDB	27.9	30.5	33.2	42.7	62.6	70.2	70.4	67.6	56.6	43.0	29.6	27.9
	2%	WB	22.6	24.7	25.3	33.0	47.3	55.3	57.4	54.3	46.0	34.0	24.6	22.0
		MCDB	24.5	27.2	27.7	37.0	58.1	66.8	68.3	62.6	52.6	36.8	26.9	23.6
	5%	WB	18.6	20.3	20.5	30.7	44.8	53.4	56.0	51.7	44.2	31.0	21.1	17.7
		MCDB	20.1	22.5	22.5	34.5	53.5	63.3	66.0	58.8	49.8	33.4	23.1	19.0
	10%	WB	14.5	16.4	15.9	27.9	42.1	51.6	54.5	49.7	42.2	28.0	16.6	13.6
		MCDB	15.9	17.9	17.6	30.8	49.8	62.0	63.6	56.4	47.1	29.8	18.0	15.1
Mean Daily Temperature Range	5% DB	MDBR	12.5	12.9	14.2	16.7	17.7	18.2	16.7	14.9	12.9	10.6	11.0	11.8
		MCDBR	15.8	14.9	16.3	16.8	24.6	24.8	24.0	21.9	19.7	11.2	11.7	16.1
		MCWBR	14.6	13.7	14.6	13.4	14.6	11.8	11.5	12.4	12.1	8.6	10.8	15.0
	5% WB	MCDBR	16.3	15.1	15.9	16.3	23.4	21.7	20.8	18.0	17.2	10.2	11.6	15.4
		MCWBR	15.2	13.8	14.2	13.0	14.3	11.0	10.7	10.7	10.8	8.2	10.8	14.5
Clear-Sky Solar Irradiance	taub	0.157	0.219	0.244	0.304	0.342	0.344	0.350	0.326	0.294	0.217	0.156	N/A	
	taud	1.909	2.108	2.239	2.117	2.163	2.382	2.402	2.473	2.449	2.212	1.917	N/A	
	Ebn at Noon	44	191	259	266	267	272	266	261	240	191	37	0	
	Edh at Noon	3	14	23	36	38	32	30	25	20	12	2	0	
All-Sky Solar Radiation	RadAvg	12	167	724	1435	1796	1762	1483	1076	637	237	34	0	
	RadStd	1	15	44	69	141	162	140	113	73	28	3	0	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	+1.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-493	N/A	N/A

Nomenclature: See separate page